

JEE MAIN REVISION MASTER: HEAT AND THERMODYNAMICS SYLLABUS REDUCED

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 28 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	First Law & Thermodynamic Processes	16 PYQs	First Law
High	Carnot Engine & Second Law Efficiency	8 PYQs	Heat Engines
High	Kinetic Theory of Gases & Thermal Radiation	8 PYQs	KTG & Radiation

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
First Law & Thermodynamic Processes	$\Delta Q = \Delta U + W$, $\Delta U = n C_v \Delta T$, $W_{\text{iso}} = nRT \ln \left(\frac{V_2}{V_1} \right)$, $W_{\text{adi}} = \frac{P_1 V_1 - P_2 V_2}{\gamma - 1}$
Carnot Engine & Second Law Efficiency	$\eta = 1 - \frac{Q_2}{Q_1} = 1 - \frac{T_2}{T_1}$, $\text{COP} = \frac{Q_2}{W} = \frac{T_2}{T_1 - T_2}$, $\beta = \frac{1 - \eta}{\eta}$
Kinetic Theory of Gases & Thermal Radiation	$v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$, $v_{\text{avg}} = \sqrt{\frac{8RT}{\pi M}}$, $\lambda_m T = b$, $E = e \sigma A T^4$, $\frac{T_1 - T_2}{t} = K \left(\frac{T_1 + T_2}{2} - T_0 \right)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 First Law & Thermodynamic Processes (Very High Yield — 16 PYQs)

Core Principle: First law of thermodynamics, molar heat capacities ($C_p - C_v = R$), and work done in cyclic/open processes.

■ **Caution / JEE Pitfall:** Internal energy $\Delta U = nC_v\Delta T$ depends ONLY on temperature change ΔT , regardless of whether the path is isobaric, isothermal, or adiabatic.

■ **Speed Trick:** In cyclic processes on a P-V diagram: clockwise cycles give positive net work (heat engine); counter-clockwise cycles give negative work (refrigerator).

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "One mole of diatomic gas having rotational modes only is kept in a cylinder with a piston system. The cross-section area of the cylinder is ..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "One mole of an ideal diatomic gas expands from volume V to $2V$ isothermally at a temperature 27°C and does W joule of work. If..." [Ans: A]
- [2026 Evening Shift] Q#2 (MCQ): "Initial pressure and volume of a monoatomic ideal gas are P and V . The change in internal energy of this gas in adiabatic expansion to ..." [Ans: C]

3.2 Carnot Engine & Second Law Efficiency (High Yield — 8 PYQs)

Core Principle: Carnot cycle efficiency limit, reversible refrigerators, and Clausius/Kelvin-Planck statements.

■ **Caution / JEE Pitfall:** Temperatures T_1 (source) and T_2 (sink) MUST strictly be substituted in Kelvin (K), never in Celsius ($^\circ\text{C}$).

■ **Speed Trick:** To maximize Carnot efficiency, lowering sink temperature T_2 by ΔT is more effective than raising source temperature T_1 by the same ΔT .

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "Two closed vessels of same volume are joined through a narrow tube and both vessels are filled with air of pressure 90 kPa and temperature 4°C ..." [Ans: A]
- [2026 Evening Shift] Q#1 (MCQ): "Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R. Assertion A : If the average kinetic en..." [Ans: C]
- [2026 Evening Shift] Q#1 (MCQ): "A mixture of carbon dioxide and oxygen has volume 8310 cm^3 , temperature 300 K , pressure 100 kPa and mass 13.2 g . The number of moles of ca..." [Ans: C]

3.3 Kinetic Theory of Gases & Thermal Radiation (High Yield — 8 PYQs)

Core Principle: Molecular speeds, energy equipartition $E = (f/2)nRT$, Stefan-Boltzmann law, Wien's displacement law, and Newton's law of cooling.

■ **Caution / JEE Pitfall:** Degrees of freedom: Monatomic $f = 3$ ($\gamma = 5/3$); Diatomic at room temp $f = 5$ ($\gamma = 7/5$); Polyatomic non-linear $f = 6$ ($\gamma = 4/3$).

■ **Speed Trick:** Net heat radiated by a body at temperature T in surroundings T_0 is $P_{\text{(net)}} = \epsilon\sigma A(T^4 - T_0^4)$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "One mole of diatomic gas having rotational modes only is kept in a cylinder with a piston system. The cross-section area of the cylinder is ..." [Ans: N/A]
- [2026 Evening Shift] Q#2 (MCQ): "An ideal gas at pressure P and temperature T is expanding such that $PT^3 = \text{constant}$. The coefficient of volume expansion of the gas i..." [Ans: C]
- [2026 Evening Shift] Q#2 (MCQ): "The temperature of a metal strip having coefficient of linear expansion α is increased from T_1 to T_2 resulting in increase ..." [Ans: D]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Morning Shift (Q#1)

Two closed vessels of same volume are joined through a narrow tube and both vessels are filled with air of pressure 90 kPa and temperature 400 K. Keeping the temperature of one vessel constant at 400 K the second vessel temperature is raised to 500 K. The final pressure in the vessels is ____ kPa.

- (A) 100 [Correct]
- (B) 120
- (C) 90
- (D) 105

Correct Answer: A

Solution Analysis: Using Ideal Gas Law: $PV = nRT$ Where P is pressure, V is volume, n is number of moles, R is the gas constant, and T is absolute temperature. Since the system is closed, the total number of moles (n_{total}) remains constant before and after heating. Initial State: Both vessels (each of volume V) are at $P_{(0)} = 90 \text{ kPa}$ and $T_{(0)} = 400 \text{ K}$.
 $n_{\text{total}} = \frac{P_{(0)}V}{RT_{(0)}} + \frac{P_{(0)}V}{RT_{(0)}} = \frac{2P_{(0)}V}{RT_{(0)}}$... (i) Final State: Pressure becomes $P_{(f)}$ in both vessels. Vessel 1 stays at $T_{(1)} = 400 \text{ K}$, and vessel 2 is at $T_{(2)} = 500 \text{ K}$.
 $n_{\text{total}} = \frac{P_{(f)}V}{RT_{(1)}} + \frac{P_{(f)}V}{RT_{(2)}}$

Benchmark Problem 2: JEE Main 2026 (Online) 23rd January Evening Shift (Q#1)

One mole of an ideal diatomic gas expands from volume V to $2V$ isothermally at a temperature 27°C and does W joule of work. If the gas undergoes same magnitude of expansion adiabatically from 27°C doing the same amount of work W , then its final temperature will be (close to) ____ $^\circ\text{C}$. ($\log_e 2 = 0.693$)

- (A) -56 [Correct]
- (B) -117
- (C) -30
- (D) -189

Correct Answer: A

Solution Analysis: Case -1 : Isothermal Expansion ($\Delta T = 0$) For an ideal gas undergoing isothermal expansion from $V_{(1)}$ to $V_{(2)}$: $W_{\text{iso}} = nRT \ln \left(\frac{V_{(2)}}{V_{(1)}} \right)$ Given : $n = 1$ mole $T = 27^\circ\text{C} = 300 \text{ K}$ $\frac{V_{(2)}}{V_{(1)}} = \frac{2V}{V} = 2$ $\ln 2 = 0.693$ $W = 1 \times R \times 300 \times 0.693 = 207.9R$ Case - 2 : Adiabatic Expansion ($\Delta Q = 0$) For an adiabatic process, the work done is given by: $W_{\text{adia}} = \frac{nR(T_{(1)} - T_{(2)})}{\gamma - 1}$ The gas is diatomic, so the adiabatic constant $\gamma = \frac{7}{5} = 1.4$. $T_{(1)} = 300 \text{ K}$ Work done is the same magnitude ($W = 207.9R$). $207.9R = \frac{nR(T_{(1)} - T_{(2)})}{\gamma - 1}$

Benchmark Problem 3: JEE Main 2026 (Online) 5th April Evening Shift (Q#2)

An ideal gas at pressure P and temperature T is expanding such that $PT^3 = \text{constant}$. The coefficient of volume expansion of the gas is ____.

- (A) $(2 / T)$
- (B) $(1 / T)$
- (C) $(4 / T)$ **[Correct]**
- (D) $(3 / T)$

Correct Answer: C

Solution Analysis: The coefficient of volume expansion (γ) is defined as the fractional change in volume per unit change in temperature at constant pressure : $\gamma = (1 / V)(dV / dT)$ The relation between pressure and temperature is, $PT^3 = C$ (where C is a constant) For an ideal gas, $PV = nRT \Rightarrow P = (nRT / V)$ Substitute P into the given process equation : $((nRT / V))T^3 = C \Rightarrow \frac{nRT^4}{V} = C \Rightarrow V = \frac{nRT^4}{C}$ Let $k = (nR / C)$ (another constant), then: $V = kT^4$ On differentiating V with respect to T : $\frac{dV}{dT} = \frac{d}{dT}(kT^4) = 4kT^3$